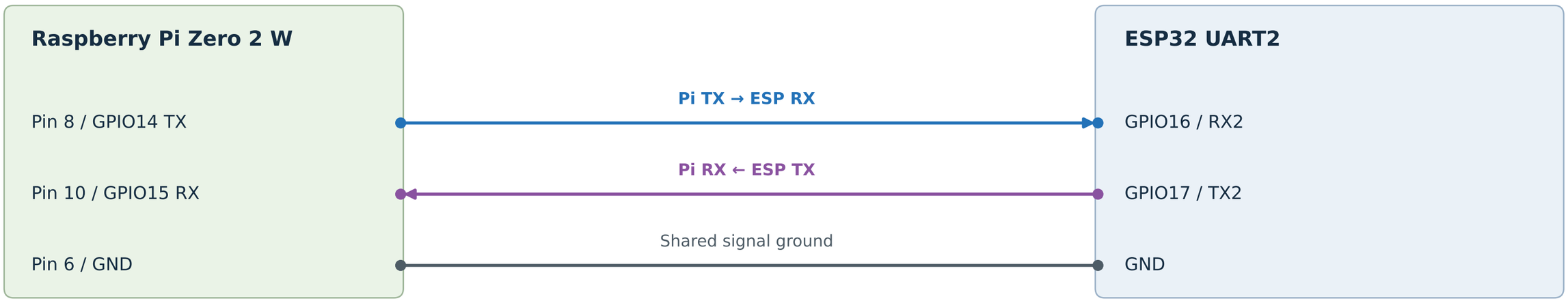
03 / LOGIC WIRING — EXACT CONNECTIONS

Classic ESP32-WROOM DevKit only. Leave the servo V+ supply physically disconnected for this sheet.

A. Pi ↔ ESP32: three signal connections

115200 baud • 8N1 • 3.3 V signals



Power Pi + ESP32 through their approved USB/5 V logic inputs. Disable Pi serial login before using the UART.

B. ESP32 I²C: connect each row as one shared electrical net

SIGNAL	ESP32 / SOURCE	PCA9685 — 0x40	MPU6050 — 0x68
SDA / data	GPIO21	SDA	SDA
SCL / clock	GPIO22	SCL	SCL
LOGIC POWER	3.3 V logic source*	VCC = 3.3 V	3.3 V-compatible input*
REFERENCE	GND	GND	GND

* Verify the actual breakout's supply labels/regulator requirements and the 3.3 V source capacity before wiring.

Keep I²C pull-ups at 3.3 V. AD0 low gives MPU6050 address 0x68. PCA9685 VCC is NOT servo V+.

No Pi SDA/SCL connection is used here. Match printed signal names on your own boards, not these drawing positions.